THE PRODUCT OF TWO REPRESENTABLE TRANSFORMATIONS IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but complete: it exhibits the factorization explicitly and names the two stability inputs, with no omission and no appeal to a single citation for the whole claim.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Composition of representable transformations is representable. Lemma lemma-composition-representable-transformations, used at proof lines 428-430. Composes the two factors of the factorization.
- Base change of a representable transformation is representable. Lemma lemma-base-change-representable-transformations, used at proof lines 425-428. Identifies each factor as a base change of a representable transformation.

Source reference. Original lemma 02WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3798–3798. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3797–3797. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let $F_i, G_i : (Sch/S)^{opp}_{fppf} \to Sets$, $i = 1, 2$. Let $a_i : F_i \to G_i$, $i = 1, 2$ be representable transformations of functors. Then*

$$a_1 \times a_2 : F_1 \times F_2 \longrightarrow G_1 \times G_2$$

is a representable transformation of functors.

Proof. Write $a_1 \times a_2$ as the composition $F_1 \times F_2 \to G_1 \times F_2 \to G_1 \times G_2$. The first arrow is the base change of a_1 by the map $G_1 \times F_2 \to G_1$, and the second arrow is the base change of a_2 by the map $G_1 \times G_2 \to G_2$. Hence this lemma is a formal consequence of Lemmas 02WA and 02WB. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.